

Veterinary Package Insert

Product Name:

SULFARIM 48% W/W POWDER

Active Ingredient:

Sulfadimidine Sodium.... 431mg
(Sulfadimidine.....400mg)
Trimethoprim.....80mg

Product Description:

A white colour and odourless powder.

Target Species:

Bovine, Equine, Poultry, Swine, Others: Horses, Calves, Pigs and Poultry

Pharmacodynamics:

Sulfadimidine is a pyrimidine sulphonamide antimicrobial agent. Sulfonamides, being structural analogues of para-aminobenzoic acid (PABA), competitively inhibit the incorporation of PABA into dihydropteroic acid, the precursor of folic acid. Subsequent reduction in the level of folic acid reduces the production of nucleic acids in sensitive bacteria. Mammalian cells require preformed folic acid and thus are unaffected by sulphonamides. The antibacterial actions of sulphonamides are reduced in the presence of blood, pus and tissue breakdown products, which contain purines and thymidine, as the bacterial requirement of folic acid is decreased in such media.

Trimethoprim is a diaminopyrimidine antimicrobial agent. It acts by preventing the reduction of dihydrofolate to tetrahydrofolate which is required by bacteria for biosynthesis of purines, pyrimidines and some amino acids.

Sulfadimidine and trimethoprim have similar antibacterial spectra, trimethoprim being approximately 20 times more potent than sulfadimidine. The combination blocks two sequential obligate enzymatic reaction in the microbial folate synthetic pathway. A synergistic action is demonstrated by the enhanced antimicrobial activity (potentiation) of the combination compared with the antimicrobial activity of either agent singly. The minimum inhibitory (MICs) of the combination for susceptible bacteria are substantially lower than those of either of the individual agents. The combination of sulphonamide/trimethoprim is bactericidal while either drug alone is bacteriostatic. Folate-synthesising bacteria which are resistant or moderately resistant to either drug alone are frequently susceptible to the combination.

The correlation of antibacterial sensitivity in vitro of in vivo is among the highest of all antimicrobial agents. The spectrum of bacteria sensitive to the combination includes: Staphylococci, Streptococci, Fusobacterium, Enterobacter, Corynebacterium (excluding Corynebacterium [Rhodococcus] equi), Salmonella, Shigella, Klebsiella, Pasteurella, Haemophilus, Proteus spp. and most E. coli, some Brucella and Nocardia spp. Most Pseudomonas spp. are insensitive.

Resistance to the combination by gram-negative bacteria is associated with the presence of R Factors. Resistance by Staphylococcus aureus and Haemophilus spp. is chromosomal, and is rarely encountered except in patients previously exposed to the combination.

The combination also effective against Pasteurella spp. implicated in pneumonia of calves and atrophic rhinitis (in combination with Bordetella bronchiseptica) in pigs, Fusobacterium necrophorum, the causative agent of calf diphtheria, enteric pathogens such as E. coli and Salmonella spp. in calves, pigs and poultry and Haemophilus gallinarum infection in poultry. Potentiated sulfanamides are also indicated in the treatment of coccidiosis in pigs and birds.

Pharmacokinetics:

Sulfadimidine is readily absorbed following oral administration and therapeutic plasma levels are rapidly attained. Sulfadimidine is widely distributed to all body tissues and fluids. Concentrations are above plasma levels in kidney, similar to plasma levels in pleural, peritoneal, synovial and ocular fluids; slightly lower is CSF, muscle and milk. Sulfadimidine undergoes extensive metabolism via acetylation, hydroxylation, oxidation and conjugation. The parent compound and its metabolites are excreted predominantly in the urine by glomerular filtration. Sulfadimidine undergoes passive tubular resorption this prolonging the time course of drug action. Plasma half-life in the horse has been variously estimated at between 9.4 and 9.8 hours.

Trimethoprim is readily absorbed from the gastrointestinal tract and, like sulfadimidine, rapidly reaches therapeutic levels in plasma. It is widely distributed in body tissues/ fluids including bone, prostate, CSF, pleura, peritoneal, synovial and ocular fluids. Studies in human and several animal species have demonstrated that trimethoprim reaches higher concentration in all tissues, except brain, than those achieved in plasma. The drug is extensively metabolised in the horse and is excreted in the urine by glomerular filtration and active tubular secretion. Plasma half-life in the horse has been variously estimated between 4.1 hours and 4.6 hours.

The elimination half-life of sulfadimidine in ruminants is influenced by the extent of drug metabolism, which is dose-related, and by renal clearance of metabolites. Although the plasma half-life of trimethoprim in ruminants is very short (1.0hour in the cow), sulphonamide-trimethoprim combinations have been shown to be efficacious when administered orally to calves. It has been demonstrated in calves that the trimethoprim content of tissues is considerably higher and more prolonged than that of serum.

Recommended Dose:

For oral use only:

1g of Sulfarim powder contains 400mg sulfadimidine and 80mg of trimethoprim.

12.5g of Sulfarim Powder contains 5g of sulfadimidine and 1g of trimethoprim.

3.13g of Sulfarim Powder contains 1.25g of sulfadimidine and 0.25g of trimethoprim.

6.26 g of Sulfarim Powder contains 2.5g of sulfadimidine and 0.50g of trimethoprim.

Horses: 12.5g of Sulfarim Powder per 200kg bodyweight twice daily, or 1 pack of 100g sulfarim powder for 1600kg bodyweight.

Calves: 3.13g of Sulfarim Powder per 50kg bodyweight daily. Administer on dampened food or mixed with honey or molasses or 1/2 pack of 100g sulfarim powder for 780kg bodyweight.

Pigs: 3.13g of Sulfarim Powder per 50kg bodyweight daily or 1/2 pack of 100g sulfarim powder for 780kg bodyweight.

Poultry: 3.13g-6.26g per 12L drinking water or 1/2 pack of 100g sulfarim powder for 188L of drinking water for 3 to 5 days.

Administer mixed thoroughly with food or in drinking water.

Mode of Administration:

Oral

Indications:

For the treatment of infections in horses, calves, pigs and poultry due to organisms susceptible to the combination of sulfadimidine and trimethoprim.

Contraindications:

Do not administer in animals with sensitivity to sulfonamides.

Environmental Properties:

Dispose of empty container by wrapping with paper and putting in garbage.

Warning and Precautions:

Risk-benefit should be considered when the following medical problems exist:

- Hepatic function impairment (delayed biotransformation may increase the risk of adverse effects)
- Renal function impairment (delayed elimination could cause accumulation of sulfonamide and metabolites, increasing the risk of adverse effects)
- Urolithiasis (sulfonamides can crystallize in the renal system under certain conditions)

Pregnancy and Lactation:

Sulfonamides and diaminopyrimidines cross the placenta in pregnant animals and some teratogenic effects have been seen with very high doses given to pregnant mice and rats.

Sulfonamides are distributed into milk, with 0.5 to 2% of the total dose found in the milk. For example, the milk-to-plasma concentration ratio for sulfadiazine and sulfadoxine was measured to be 0.5 in cows. Trimethoprim is distributed into milk. Trimethoprim concentrations in milk were found to be 1.3 to 3.5 times the plasma concentration measured at the same time in goats. The concentration of trimethoprim in the milk of cows is 1 to 3 times higher than in plasma and the concentration of trimethoprim in the milk of pigs is 1.3 to 3.5 times higher than in plasma.

Interactions with Other Medicaments

Sulfonamides may interact with other drugs, including warfarin, methenamine, dapsone, and etodolac. They may potentiate adverse effects caused by methotrexate and pyrimethamine. Sulfonamides will increase metabolism of cyclosporine resulting in decreased plasma concentrations. Methenamine is metabolized to formaldehyde, which may form a complex and precipitate with sulfonamides. Sulfonamides administered to horses that are receiving detomidine may develop cardiac arrhythmias. This precaution is only listed for intravenous forms of trimethoprim-sulfonamides.

Symptoms and Treatment of Overdose:

Several adverse effects have been reported after prolonged treatment, including bone marrow depression (aplastic anemia, granulocytopenia, thrombocytopenia), hepatitis and icterus, peripheral neuritis and myelin degeneration in the spinal cord and peripheral nerves, photosensitization, stomatitis, conjunctivitis, and keratitis sicca. It can lead to decreased egg production and growth. Up to 10 times the recommended dose of trimethoprim has been given with no adverse effects. Prolonged administration of trimethoprim at reasonably high concentrations leads to maturation defects in hematopoiesis due to impaired folinic acid synthesis. This effect is readily reversible by supplementation with folinic acid.

Adverse Effects:

Adverse effects associated with sulfonamides include allergic reactions, Type II and Type III hypersensitivity, arthropathy, anemia, thrombocytopenia, hepatopathy, hypothyroidism (with prolonged therapy), keratoconjunctivitis sicca, and skin reactions. Dogs may be more sensitive to sulfonamides than other animals because dogs lack the ability to acetylate sulfonamides to metabolites. Other, more toxic metabolites may persist.

Withdrawal Period:

Meat: Calves, Pig, Poultry – Do not use less than 14 days before slaughter for human consumption.

Do not use in laying birds where eggs or egg products are to be used for human consumption or processing.

Horse: 28 days

Packaging:

Packed in 100gm, 500gm, and 1kg aluminium foil

Storage Condition:

Store below 30°C in dry place, protected from light.

Shelf Life

Two years from date of manufacturing

To be used within 24 hours after first opening of container.

To be used within 24 hours after dilution.

Product Registration Holder

DEXCHEM INDUSTRIES SDN. BHD.

(Company No.: 555717-U)

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